Reg. No. | | | | | | | | | | | [14AE36]

## THIRD SEMESTER B. Tech EXAMINATION, DECEMBER – 2015

## ENGINEERING MECHANICS

**Time: 3 Hours** **Maximum Marks:70**

**Instructions:** Answer any **FIVE** full Questions. Take at least TWO questions from each part.

### PART – A

**1.** **a.** A rod of length 'L' tapers uniformly from a diameter 'D' at one end to a diameter 'd' at the other. Derive the expression for the extension(deformation) caused by an axial load P. **07**

**b.** Define:

**i.** Modulus of Rigidity **ii.** Young's modulus

**iii.** Volumetric strain **iv.** Thermal stress

Also draw the stress strain curve for a ductile material. **06**

**c.** Choose the correct one:

Ductility of a material is best measured by

**i.** Percentage elongation

**ii.** Reduction in area or contraction

**iii.** Both i and ii

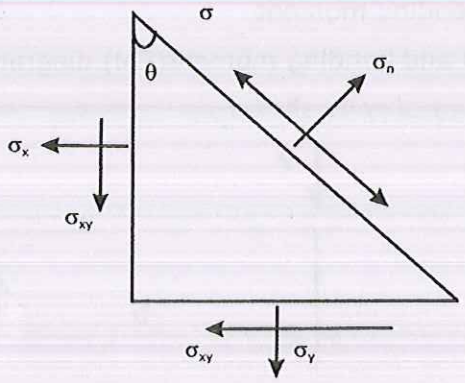
**iv.** None of the above **01**

**2.** **a.** Define principal planes and principal stresses. What is a Mohr's circle? **03**

**b.** The state of stress at a certain point in a strained material is shown in figure below. Calculate:

**i.** Normal stress, shear and resultant stress or the plane

**ii.** Principal stresses and inclination of the principal planes **10**

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**c.** Choose the correct Statement:

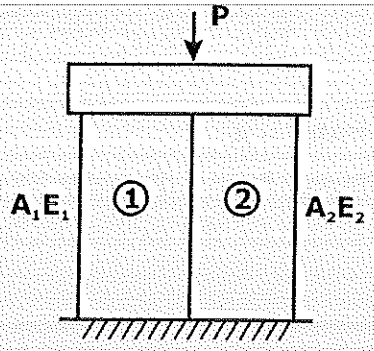
**i.** There is a single-angle relationship between Mohr's circle and real space

**ii.** There is a double-angle relationship between Mohr's circle and real space

**iii.** Both i and ii are correct

iv. None of the above **01**

**3.** **a.** A compound bar is made up of rod area $A_1$ and modulus $E_1$, and tube of equal length of area $A_2$, and modulus $E_2$. If a compressive load P is a applied to the compound bar, find how the load is shared: **04**




**b.** Define thermal strain. A steel Plate ¼ in x 2 in is 12 in long. It is placed into a 12in opening where it exactly fits. Assuming that the opening does not change in size with temperature, find the load and stress in the plate when the temperature is increased 120°F. Take coefficient of thermal expansion of steel is $6.5\times10^{-6}$ per °F and $E=30\times10^{6}$psi. **09**

**c.** The relation between modulus of Elasticity (E), Modulus of Rigidity(G) and Bulk modulus(k) is given by,

i. $E=\dfrac{3KG}{G+9k}$ ii. $E=\dfrac{9KG}{G+3k}$

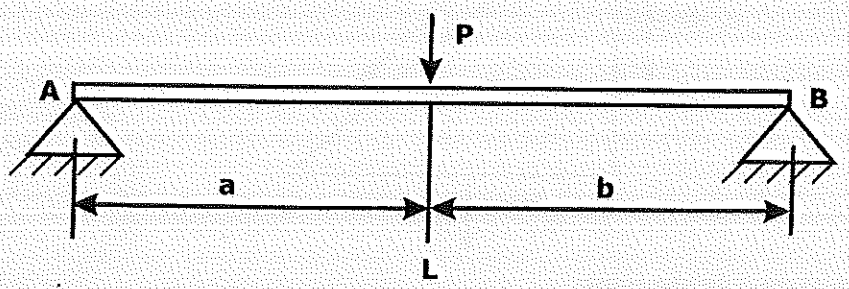
iii. $E=\dfrac{G+9k}{3KG}$ iv. $E=\dfrac{G+3k}{9KG}$ **01**

**4.** **a.** What is shear force diagram? What do you mean by point of contraflexure? Write the relation between:

i. Load and shear force

ii. Shear force and bending moment **04**

**b.** Draw the shear force(SF) and bending moment(BM) diagrams for the loading shown below. Also locate the point of contraflexure. **09**




**c.** IF the shear force across a section of a beam behaves linearly the bending moment in that section of the beam will behave:

i. Parabolically ii. Linearly

iii. Cubically iv. None of the above **01**

## PART – B

**5.** **a.** List the assumptions made in the theory of simple bending. Derive the relation $\frac{M}{I}=\frac{\sigma}{y}=\frac{E}{R}$ where the symbols have their usual meanings. **10**

**b.** Calculate the maximum stress induced in a cast iron pipe of external diameter 40mm, internal diameter 20mm and of length 4m when the pipe is supported in its ends and carries a point load of 80N at its Centre. **03**

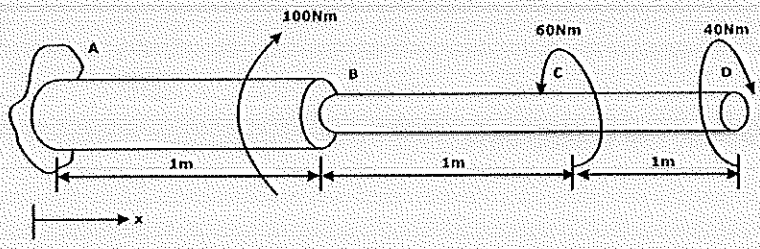
**c.** The ratio of depth to width of the strongest beam that can be cut from a circular log of diameter 'd' is,

i. $\sqrt{2}$ ii. $1/\sqrt{2}$

iii. 2 iv. $1/2$ **01**

**6.** **a.** Explain with a neat diagram the difference between bending moment and torque. **03**

**b.** For the shaft subjected to the applied torques as shown, determine the absolute maximum shear stress. Radius of AB is 80mm and radius of BC and BD are 40mm. **10**




**c.** Choose the correct one:

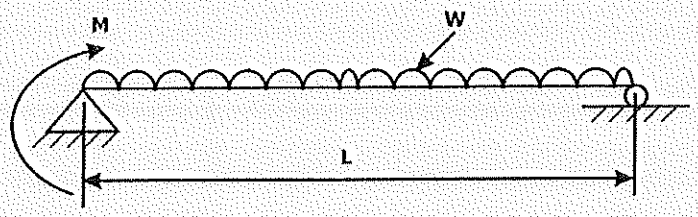
Two shafts, one solid and the other hollow, are made of the same materials and are saving same length and weight. The hollow shaft as compared to solid shaft is:

i. More strong ii. Less Strong

iii. Having same strength iv. None of the above **01**

**7.** **a.** List the methods used for determining slope and deflection at a section in a loaded beam. State their usages. **03**

**b.** Find the slope and deflection equation for the beam shown in figure below by double integration method. Here, M=14.4K Nm, w=200N/m, L=12m. **10**




**c.** Choose the correct one:

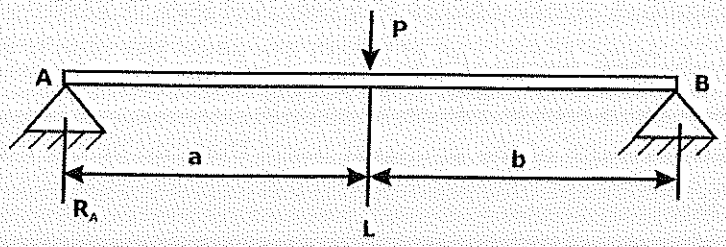
The expression $EI\frac{d^2y}{dx^2}$ at a section of a member represents:

i. Shearing force ii. Rate of loading

iii. Bending moment iv. Slope **01**

**8.** **a.** State Engessor's and Maxwell's reciprocal theorem. **04**

**b.** Determine the expression for strain energy of the prismatic beam AB for the loading as shown in figure below. Take into account only the effect of normal stresses due to bending: **09**




Evaluate the strain energy of the following values of the beam:

P=208kN; L=3.6m; Q=0.9m; b=2.7m, E=200Gp and $I=104\times10^8 mm^4$.

**c.** Choose the correct one:

The proof resistance is given by

i. $\frac{2\sigma^2}{E}$ ii. $\frac{2E}{\sigma^2}$

iii. $\frac{\sigma^2}{2E}$ iv. $\frac{E}{2\sigma^2}$ **01**

Where $\sigma$ =stress at the elastic limit.